END-TO-END DATA ANALYTICS PROJECT

Retail Customer Behaviour & Shopping Trends

Python | PostgreSQL | Power BI | Business Insights

PORTFOLIO PRESENTATION

Prepared using the project brief and the Power BI dashboard screenshot supplied for this presentation.

1. Project Overview

This project analyses retail customer behaviour and shopping trends to turn transaction data into useful business information. It focuses on customer spending, product performance, discounts, subscriptions, shipping preferences and seasonal revenue.

Business objective

Help a retail business understand who its customers are, what they purchase, how much they spend and which factors are associated with purchasing patterns.

Project goals

- Prepare and explore the dataset
- Answer business questions using SQL
- Build an interactive Power BI dashboard
- Communicate verified findings and recommendations

2. Dataset & Analysis Questions

Dataset: Retail Customer Behaviour / Shopping Trends. Fields used include CustomerID, Age, Gender, Item Purchased, Category, Purchase Amount (USD), Review Rating, Discount Applied, Subscription Status, Shipping Type, Previous Purchases and Season.

Business questions

- Which discount-using customers spend above average?
- Which products have the highest average review ratings?
- How do Standard and Express shipping purchase amounts compare?
- Do subscribers spend more than non-subscribers?
- Which products have the highest discounted-purchase percentage?
- How can customers be segmented as New, Returning and Loyal?
- What are the top three products in each category?
- Which age group has the highest average purchase amount?
- Which season generates the highest revenue?

Customer segment thresholds should be documented consistently in the final report.

3. End-to-End Workflow

Stage	Tool	Work completed / planned
Dataset review	Excel	Understand columns, data types and data quality.
Cleaning & EDA	Python / pandas	Inspect missing values, duplicates, distributions and patterns.
Business analysis	PostgreSQL	Use filtering, aggregation, CASE, subqueries and window functions.
Visualisation	Power BI	Create KPI cards, charts and interactive slicers.
Documentation	PDF report	Explain methods, verified findings and recommendations.
Portfolio	GitHub / LinkedIn	Publish organised project files and a concise project summary.

4. Python – Data Preparation & EDA

Python and pandas are used to load the source dataset, inspect its structure and prepare it for analysis.

Analysis steps

- Load the CSV dataset and inspect rows, columns and data types.
- Check missing values and duplicate records.
- Validate numeric fields such as Age, Purchase Amount and Review Rating.
- Review category values such as Season, Shipping Type and Subscription Status.
- Explore purchase amounts, age groups, products and categories using summary statistics and charts.

Evidence to include: notebook screenshots showing data-quality checks, cleaning decisions and key exploratory charts.

5. PostgreSQL – Business Analysis

PostgreSQL is used to answer business questions with SQL queries and compare customer and product behaviour.

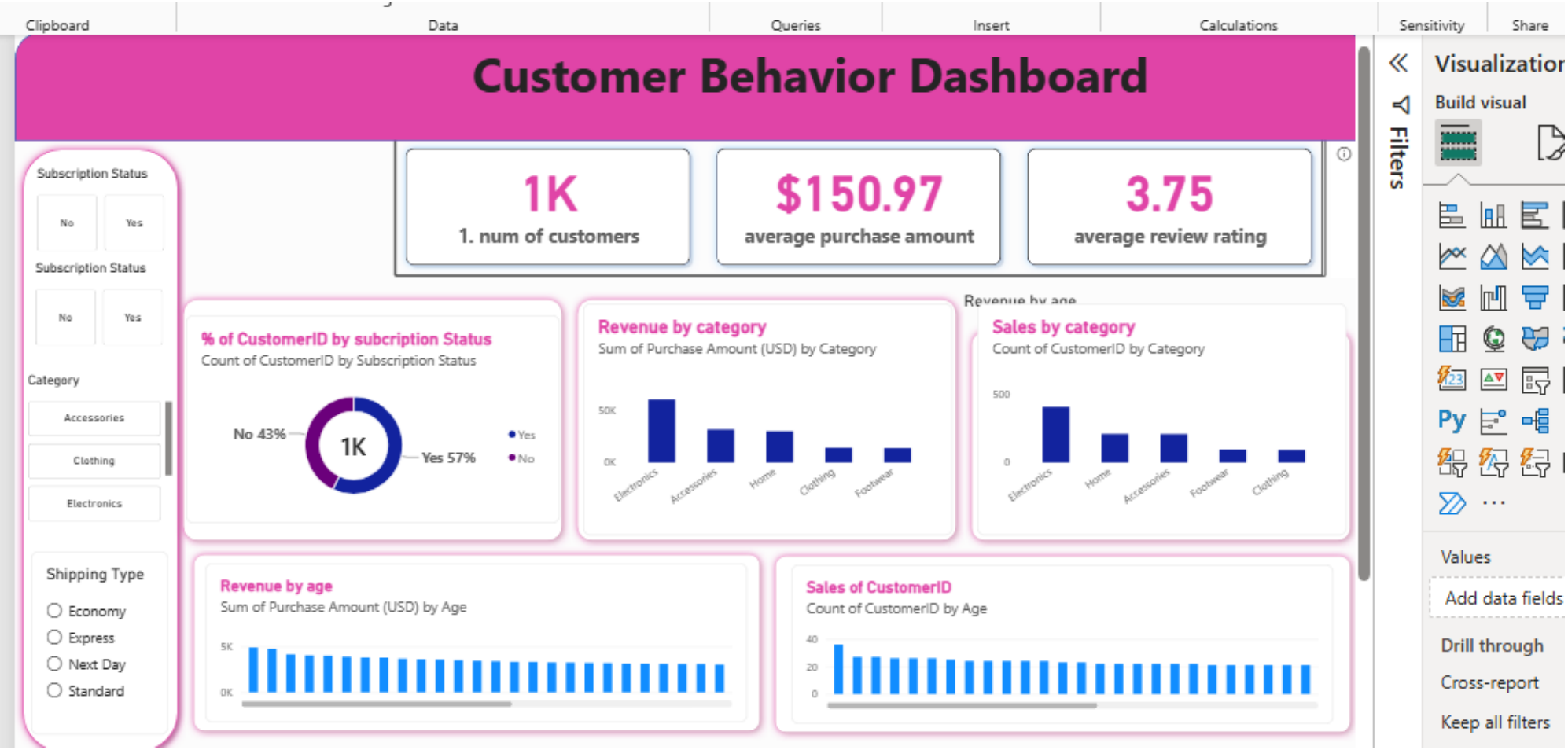
SQL techniques

- WHERE and subqueries – identify discount users spending above average.
- AVG and GROUP BY – compare average ratings and purchase amounts.
- SUM and CASE – calculate revenue and discount percentages.
- CASE expressions – create customer segments from previous purchases.
- DENSE_RANK with PARTITION BY – rank top products within each category.

PostgreSQL note: column names containing spaces or special characters must be enclosed in double quotes, for example "Purchase Amount (USD)".

6. Power BI – Customer Behaviour Dashboard

The dashboard combines KPI cards, customer subscription analysis, category-level revenue and sales, and age-based purchase trends. Slicers allow users to explore the report by subscription status, category and shipping type.



Dashboard screenshot supplied for this presentation. Validate all labels and figures in the final PBIX before submission.

7. Dashboard Walkthrough

1K	$150.97	3.75
Customer count	Average purchase amount	Average review rating

Visuals shown in the supplied screenshot

- Donut chart – customer count by subscription status.
- Revenue by category – sum of purchase amount by product category.
- Sales by category – customer count by category.
- Revenue by age – purchase amount grouped by age.
- Customer count by age – customer count grouped by age.
- Slicers – subscription status, category and shipping type.

The displayed KPI values are transcribed from the supplied screenshot; confirm the final figures and aggregation logic in Power BI.

8. Findings & Business Recommendations

Use the completed SQL outputs and Power BI filters to enter evidence-based findings. Do not add conclusions until the results have been checked.

Findings to record

Topic	Result to add
Discount behaviour	Number of discount-using customers spending above the overall average.
Product ratings	Top five products and their average review ratings.
Shipping	Standard and Express average purchase amounts.
Subscription	Average spend and total revenue for subscribers vs non-subscribers.
Discount rate	Five products with the highest percentage of discounted purchases.
Segments / season	Customer segment counts and highest-revenue season.

Recommendations should follow the evidence: for example, targeted promotions, customer-retention campaigns or product assortment changes only where supported by the analysis.

9. GitHub Portfolio & Deliverables

Suggested repository structure

Retail-Customer-Behaviour-Analysis/
■■■ data/ – dataset (only if sharing is permitted)
■■■ python/ – cleaning and EDA notebook
■■■ sql/ – PostgreSQL queries
■■■ powerbi/ – PBIX dashboard
■■■ report/ – final project report
■■■ presentation/ – presentation PDF
■■■ README.md – project overview and instructions

Publishing checklist

- Remove passwords and private connection details.
- Include only data you are permitted to share.
- Add screenshots of the dashboard and important SQL outputs.
- Explain the tools, workflow and your contribution.
- Check that every metric in the report matches the final analysis.

10. Conclusion & Thank You

Retail Customer Behaviour & Shopping Trends

This project demonstrates an end-to-end analytics workflow: understanding data, preparing it with Python, answering business questions using PostgreSQL, and communicating patterns through an interactive Power BI dashboard.

Portfolio details

Name: _____
Course / Institution: _____
GitHub repository: _____
LinkedIn profile: _____

From data to decisions — a complete portfolio analytics project.